SS COOLING TOWER CONSULTANTS

Thermal Design · CT Assessment & Upgrade · CT Testing · Technical Management & Training

Test

POST TEST — SINGLE CELL AFTER DISTRIBUTION SYSTEM & FAN CHANGE

Test

OWNER / CLIENT	**Test**
TESTING AGENCY & EQUIPMENT PROVIDER	**SS COOLING TOWER CONSULTANTS (SSCTC)**
DATE OF TEST	**Test**
REPORT DATE	**Test**
ASSESSMENT METHOD	**CTI ATC-105 Merkel Method**

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PREAMBLE

p1

MEMBERS PRESENT FOR THE TEST

CLIENT — Test

- m1

SS COOLING TOWER CONSULTANTS (SSCTC)

- m2

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CONCLUSION AND REMARKS

1. c1

SUGGESTIONS AND SPECIAL NOTES

1. s1

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AIR FLOW DATA — ANEMOMETER TRAVERSES (CTI ATC-143)

Metric	Value	Unit
Average Measured Velocity	4.99	m/s
Total Free Area	92.25	m^2
Total Air Flow	**460**	**m^3/s**

Air flow measured per CTI ATC-143 Method of Equal Area — 10 traverses per quadrant (40 readings total for one fan).

ATC-105 PERFORMANCE EVALUATION — CTI METHOD

Parameter	Unit	Design Values	Recorded Test
Water Flow Rate	M^3/hr	3863.6	3680.0
Hot Water Temperature	Deg C	43.0	42.13
Cold Water Temperature	Deg C	33.0	32.4
Wet Bulb Temperature	Deg C	29.0	21.7
Range	Deg C	10.0	9.73

STEP 1

Table 1 — CWT vs Range at Test WBT = 9.73°C | (Design Range = 10.0°C)

3 performance curves at 90%, 100% and 110% water flow. A vertical line is drawn at the test WBT to intersect each of the 3 Range conditions. CWT values are read at each intersection.

Range (%)	Range (°C)	CWT @ 90% Flow (3477 m³/hr)	CWT @ 100% Flow (3864 m³/hr)	CWT @ 110% Flow (4250 m³/hr)
80%	8.0 °C	27.41	28.44	29.43
100%	**10.0 °C**	**27.96**	**29.11**	**30.21**
120%	12.0 °C	28.33	29.56	30.73

STEP 2

Table 2 — CWT vs Water Flow at Test Range = 9.73°C (Cross Plot 1 intersections)

From Cross Plot 1 — draw a vertical line at the test range to read CWT at each of the three water flows.

Flow Condition	Water Flow (m³/hr)	CWT at Test Range (°C)
90% Flow (Purple)	3477.24	27.89
100% Flow (Green)	3863.6	29.023
110% Flow (Blue)	4249.96	30.105

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STEP 2 — CROSS PLOT 1

CWT vs RANGE @ Test WBT — Three Water Flow Curves with Test Range Intersection

Plot CWT (Y-axis) vs Range (X-axis) for 90%, 100% and 110% water flows. Draw a vertical line (red) at the test range of 9.73°C. The intersection points give the Table 2 values used in Cross Plot 2.

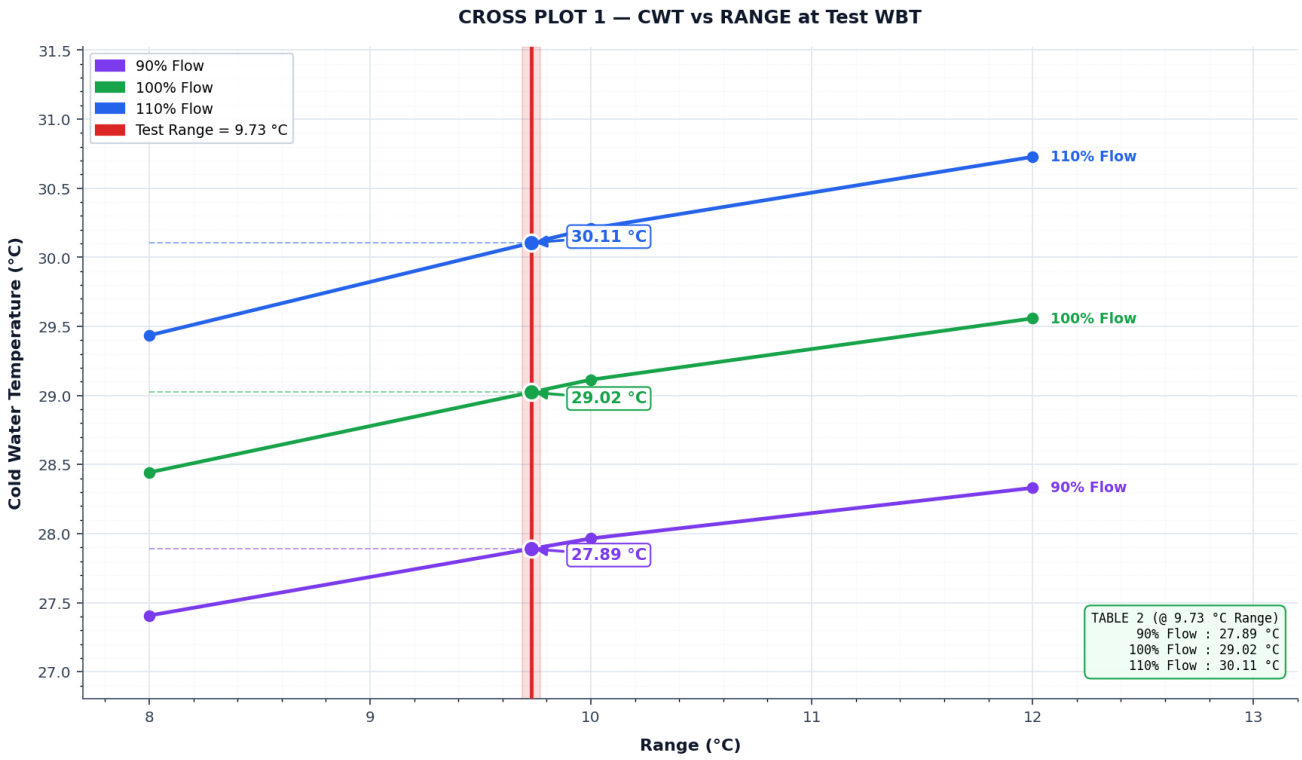

FIGURE 1 — Cross Plot 1: CWT vs RANGE at Test WBT = 21.7°C | Test Range (vertical red line) = 9.73°C

STEP 3

Plot Cross Plot 2 — Water Flow (X) vs CWT (Y) from Table 2 data

Using the three data points from Table 2, plot CWT on Y-axis and Water Flow on X-axis. Extend the curve to cover the adjusted water flow range.

STEP 4

Calculation of Adjusted Water Flow (Q_{adj})

Step 4 Input Parameters

Ref	Parameter	Value	Unit
a	**Fan Power Design (W_d)**	117	KW
b	**Fan Power Test (W_t)**	117	KW
c	**Hot Water Temp. (Test)**	32.4	Deg C
d	**Cold Water Temp. (Test)**	32.4	Deg C
e	**Range (Test)**	9.73	Deg K
f	**WBT (Test)**	21.7	Deg C
h	**Test Flow (Q_{wt})**	3477	M³/hr
i	**Density Ratio ($\rho_{test}/\rho_{design}$)**	1.0337	—

Adjusted Flow $Q_{adj} = Q_{wt} \times (W_d/W_t)^{■} \times \rho^{■}$	**3720.88**	**m³/hr**

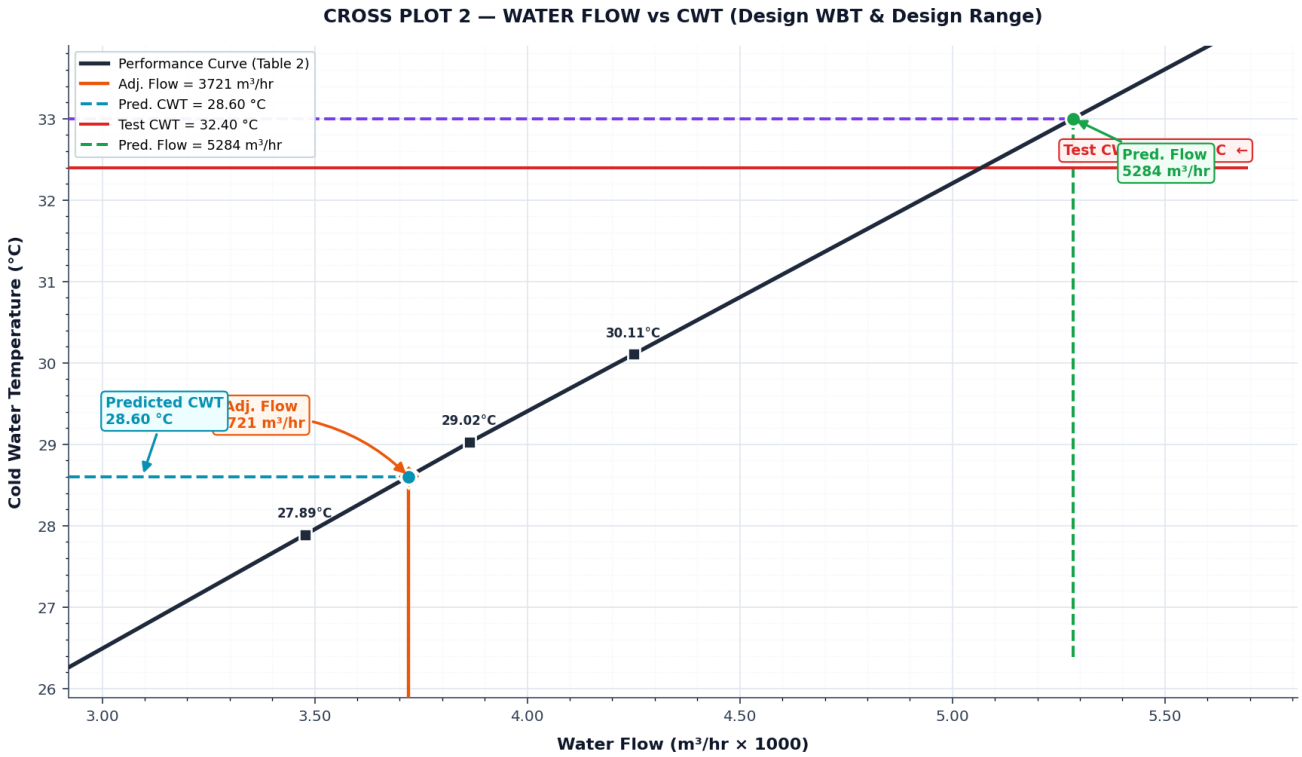

FIGURE 2 — Cross Plot 2: Water Flow vs CWT @ Design WBT & Design Range | Adj. Flow = 3720.88 m³/hr

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FINAL ATC-105 MATHEMATICAL RESULTS

STEP 5

Predicted CWT, Performance Shortfall and Cooling Tower Capability

Ref	Item	Value	Unit
i	**Adjusted Water Flow in Cross Plot 2 X-axis**	3720.88	m³/hr
ii	**From adj. flow draw vertical line → intersect curve → read CWT**	(see Cross Plot 2, orange vertical line)	
iii	**Predicted (Expected) Cold Water Temperature**	**28.60**	**Deg C**
iv	**Tolerance on CWT (Specification)**	0.00	Deg C
v	**Total Predicted CWT (incl. tolerance)**	**28.60**	**Deg C**
vi	**Test Cold Water Temperature (Recorded site data)**	32.4	Deg C

vii	**Predicted Water Flow (at Design CWT on Cross Plot 2)**	5283.71	m³/hr

CONCLUSION

3.80 °C

Cooling Tower Shortfall in Performance

Test CWT (32.4°C) is higher than Predicted CWT (28.60°C) by 3.80°C

70.4 %

COOLING TOWER CAPABILITY

Capability (%) = (Adjusted Water Flow / Predicted Water Flow) × 100
= 3720.88 / 5283.71 × 100

∴ Hence cooling tower shortfall in performance is 3.80 °C | Capability = 70.4 %

In the above case, the "Test Cold Water" is higher than "Predicted Cold Water" by 3.80°C. Mark the Adjusted Water Flow in X-axis and project the line to intersect the Flow vs CWT graph (Cross Plot 2). Capability (%) = (Adjusted Water Flow / Predicted Water Flow) × 100.